



Divit Mittal

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WORK EXPERIENCE

REMOTE

OPEN SOURCE & COMMUNITY CONTRIBUTIONS – 01/07/2024 – CURRENT

Active contributor to the functional programming ecosystem, particularly within the Nix community. Participated in Nix consulting bounties with **Tweag**, **ModusCreate**, and **Blockfrost**, providing expertise on complex NixOS configurations, package derivations, and infrastructure automation for enterprise clients.

UBIOPS (YES!DELFT) – DELFT, NETHERLANDS

DEVOPS & SYSTEM ADMINISTRATOR – 15/12/2021 – 28/02/2022

Administered containerized environments using Docker/Kubernetes for multi-tenant ML workloads, implemented CI/CD pipelines for automated model deployment, and maintained cloud infrastructure across AWS/GCP.

EDUCATION AND TRAINING

10/09/2022 – CURRENT Jaipur, Rajasthan, India

BACHELOR OF TECHNOLOGY IN DATA SCIENCE & ENGINEERING Manipal University

Website <https://jaipur.manipal.edu/> | National classification NCC5

01/09/2021 – 01/09/2022 Delft, Netherlands

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING Delft University of Technology

Website tudelft.nl | Final grade 8.0 | Type of credits ECTS | Number of credits 20

01/08/2020 – 28/08/2020 Singapore, Singapore

BACHELOR OF COMPUTING IN COMPUTER SCIENCE National University of Singapore

Website <https://www.nus.edu.sg/>

31/03/2018 – 12/07/2020 New Delhi, India

ALL INDIA SENIOR SCHOOL CERTIFICATE EXAMINATION Central Board of Secondary Education (CBSE)

Address Delhi Public School, R.K. Puram, New Delhi, India | Final grade A1 : ~96%

31/03/2016 – 28/05/2018 New Delhi, India

ALL INDIA SECONDARY SCHOOL EXAMINATION Central Board of Secondary Education (CBSE)

Address Bhatnagar International School, Bhera Enclave, Paschim Vihar, New Delhi, India | Final grade A1 : ~97%

LANGUAGE SKILLS

Mother tongue(s): **HINDI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
GERMAN/DEUTSCH (GOETHE- ZERTIFIKAT B2)	B2	B2	B2	B2	B2
ENGLISH (IELTS)	C2	C2	C1	C2	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Machine Learning & Data Science (Mathematics)

Deep Learning: Keras & PyTorch | Machine Learning: scikit-learn, tensorflow | LLM: Transformers, RAG, Encoders, & Decoders | Big Data: Hadoop, Spark, Hive, Kafka | Computer Vision: OpenCV, ITK

DevOps & Linux SysAdmin

NixOS Expert | IaC: nix, ansible & terraform | VCS: Git & Hg | Cloud Experience: AWS, GCP & Firebase | Shells: bash, zsh, fish, pwsh | Virtualization & Containerization: Proxmox, KVM, Docker & Podman | CI/CD: GitHub Actions & Jenkins

PROJECTS

04/2024 – CURRENT

OS-nixCfg: A Personal Infrastructure as Code configuration

Nix, nix-darwin, **NixOS**, home-manager, etc. configurations to attain my **deterministically reproducible** OS, written in the **functional** "Nix" expression language. Maintains:

- Darwin-kernel systems, i.e. macOS
- Android systems
- Desktop Linux-kernel systems, e.g. NixOS, Arch Linux, RHEL
- WSL2 on Windows

Link <https://github.com/DivitMittal/OS-nixCfg>

02/2024

Programming Languages Documentation Conversational AI @ IEEE MUJ

RAG-enhanced **LLaMa-like transformer** neural network based LLM for assistance on programming languages/frameworks' documentations, with a **streamlit** front-end. Everything was implemented from scratch:

- Pre-trained **LLaMa-like transformer** with capability from **tokenization** to **embedding**(positional & word2vector) to output probabilities.
- **RAG** was implemented via **Qdrant** vector database on a **DigitalOcean Docker** container.

Link <https://github.com/DivitMittal/DocAssist-LLM>

10/2023

Generation of "Hazard Map" for safely navigating a lunar lander

Is implemented by utilising the sensory data from the **ISRO's** Chandrayaan-2 TMC2 & OHRC payloads for **SIH**(Smart India Hackathon). The project entails wrangling multiple data in form of multispectral high-res. imagery & stereo-triplet panchromatic data of lunar terrain. These multiple streams of data are fused together and analysed respectively for ground truth via **SRGANs** & **DCNNs**.

HONOURS AND AWARDS

09/2024

BobbleAI Datathon Winner – BobbleAI Keyboard, Gurgaon, Haryana

Processed raw product data, i.e. [BobbleAI Keyboard](#)(7Cr+ users) event records in the last 5 days before July 1, 2024,i.e., about 8 million rows. Extracted & transformed JSON, to preprocess, ML-filter and store as structured **Apache Parquet** on **AWS S3** via serverless **AWS Lambda** functions & **PySpark** on **AWS Glue**.

10/2023

Proglint's Computer Vision 2k23 National Hackathon 5th Position Holder – Alliance University, Anekal, Karnataka

- Leverage the high-fidelity **CARLA** simulator to create diverse driving scenarios for an **autonomous vehicle driving system** with RGBA & LiDAR sensors. Utilizing deep learning techniques, i.e. **CNNs** for **semantic segmentation** to improve the vehicle's understanding of its surroundings, & manoeuvre.
- A **Driver Drowsiness Detection** system that uses **Siamese Neural Networks** and computer vision techniques, like **haarcascade** wave filters to monitor a driver's facial features in real-time and detect signs of fatigue. By analyzing eye states, & mouth movements, the system identifies potential drowsiness and enhances road safety.

Link <https://github.com/DivitMittal/CARLA-Autonomous-Driving>

01/2020

International Rank 7 at the International Mathematics Olympiad – The Science Olympiad Foundation, New Delhi

12/2019

International Rank 49 at the International English Olympiad (IEO) – The British Council Division, British High Commission, New Delhi

01/2019

India National Level Runner-up at the Vidyarthi Vigyan Manthan – Vijnana Bharati (VIBHA), with Vigyan Prasar, under the Department of Science and Technology & NCERT
